

Calc 1 Lecture 20: More curve sketching...

Information that can be used to sketch a graph of a function f (without a graphing calculator):

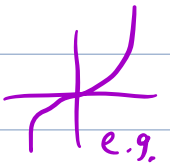
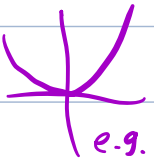
A. Domain where is the function defined

B. Intercepts

- x intercepts ($y=0$)
- y intercept ($x=0$)

C. Symmetry

- even function ($f(-x) = f(x)$)
symmetric about y axis
- odd functions ($f(-x) = -f(x)$)
symmetric about origin
- periodic functions ($f(x+k) = f(x)$)
period



D. Asymptotes

- vertical asymptotes
— if $x=a$ is VA, evaluate $\lim_{x \rightarrow a^-} f(x)$, $\lim_{x \rightarrow a^+} f(x)$
- horizontal asymptotes
— $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$

E. Intervals of Increase or Decrease

↓ • sign analysis on f' ($f' > 0$: increase)
($f' < 0$: decrease)

F. Local Max and Mins

• applying 1st deriv. test or 2nd deriv. test at critical points

G. Concavity and points of inflection • sign analysis on f''

⇒ sketch!

($f'' > 0$: concave up)
($f'' < 0$: concave down)

Example sketch $y = \frac{2x^2}{x^2-1} = f(x)$ using the above guidelines.

A. Domain: $x^2 - 1 \neq 0$ (no division by 0)

$$x^2 \neq 1 \Rightarrow x \neq \pm 1$$

$$\text{domain: } (-\infty, -1) \cup (-1, 1) \cup (1, \infty)$$

B. intercepts:

$$y\text{-intercept: } x=0: y = \frac{2 \cdot 0^2}{0^2 - 1} = \boxed{0}$$

$$x\text{-intercepts: } y=0: 0 = \frac{2x^2}{x^2-1} \Rightarrow 2x^2=0 \Rightarrow \boxed{x=0}$$

C. symmetry? $\frac{2x^2}{x^2-1} = f(x)$

even? $f(-x) = f(x)$?

$$f(-x) = \frac{2(-x)^2}{(-x)^2 - 1} = \frac{2x^2}{x^2 - 1} = f(x) \quad \checkmark \quad \text{yes}$$

periodic? no

D. Asymptotes

vertical asymptotes: division by 0 $\Rightarrow \underline{x = -1, \text{ or } x = 1}$

sign analysis

$$\bullet \lim_{x \rightarrow -1^-} \frac{2x^2}{x^2-1} \begin{matrix} \text{Pos} \\ \text{Pos} \end{matrix} = \infty$$

$$\bullet \lim_{x \rightarrow -1^+} \frac{2x^2}{x^2-1} \begin{matrix} \text{Pos} \\ \text{neg} \end{matrix} = -\infty$$

$$\bullet \lim_{x \rightarrow 1^-} \frac{2x^2 \text{] pos}}{x^2 - 1 \text{] neg}} = -\infty$$

$$\bullet \lim_{x \rightarrow 1^+} \frac{2x^2 \text{] pos}}{x^2 - 1 \text{] pos}} = \infty$$

$$x^2 - 1 \stackrel{?}{>} 0 \Leftrightarrow x^2 > 1 \Leftrightarrow x > 1 \text{ or } \underline{x < -1}$$

Horizontal asymptotes

$$\lim_{x \rightarrow \infty} \frac{2x^2}{x^2 - 1}$$

approach 1: $\lim_{x \rightarrow \infty} \frac{2x^2}{x^2 - 1} = \lim_{x \rightarrow \infty} \frac{\frac{2x^2}{x^2}}{\frac{x^2}{x^2} - \frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{2}{1 - \frac{1}{x^2}} \rightarrow 0$
 mult. by, divide by dominant term
 $= \frac{2}{1} = \boxed{2}$

approach 2: L'Hospital's

$$\lim_{x \rightarrow \infty} \frac{2x^2}{x^2 - 1} = \lim_{x \rightarrow \infty} \frac{4x}{2x} = \boxed{2}$$

$$\lim_{x \rightarrow -\infty} \frac{2x^2}{x^2 - 1} = \boxed{2} \quad (\text{same reasoning})$$

E increase / decrease

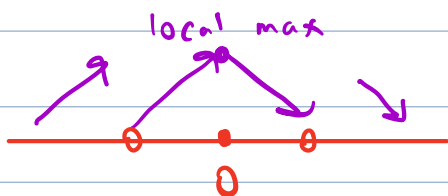
$$f(x) = \frac{2x^2}{x^2 - 1} \Rightarrow f'(x) = \frac{\text{quotient rule} \quad 4x(x^2 - 1) - 2x^2(2x)}{(x^2 - 1)^2}$$

$$= \frac{\cancel{4x^3} - 4x - \cancel{4x^3}}{(x^2 - 1)^2} = \frac{-4x}{(x^2 - 1)^2}$$

Sign analysis: $\frac{-4x}{(x^2-1)^2}$ } pos if $x < 0$
neg if $x > 0$

$f' > 0$ if $x < 0$: $(-\infty, -1) \cup (-1, 0)$ $\leftarrow \pm 1$ not in domain
 $f' < 0$ if $x > 0$: $(0, 1) \cup (1, \infty)$

F. local min/max



critical points:

$$f'(x) = \frac{-4x}{(x^2-1)^2} = 0 \Rightarrow$$

$$x = 0.$$

local max

$$f'(x) \text{ undefined} \Rightarrow x^2 - 1 = 0$$

$$\Rightarrow x = \pm 1$$

not in domain, don't count

G. Concavity and inflection points analyze f''

$$f(x) = \frac{2x^2}{x^2-1}$$

$$f'(x) = \frac{-4x}{(x^2-1)^2} = -4 \frac{x}{(x^2-1)^2}$$

$$f''(x) = -4 \frac{d}{dx} \left(\frac{x}{(x^2-1)^2} \right) \quad \text{quotient rule}$$

$$= -4 \cdot \frac{\overbrace{\frac{d}{dx}(x)}^1 (x^2-1)^2 - x \cdot \overbrace{\frac{d}{dx}((x^2-1)^2)}^{2(x^2-1) \cdot 2x}}{(x^2-1)^4}$$

$$= -4 \cdot \frac{(x^2-1)^2 - 4x^2(x^2-1)}{(x^2-1)^4}$$

$$= -4 \cdot \frac{\cancel{(x^2-1)} ((x^2-1) - 4x^2)}{\cancel{(x^2-1)} (x^2-1)^3}$$

$$= -4 \cdot \frac{-3x^2-1}{(x^2-1)^3} = \boxed{4 \frac{3x^2+1}{(x^2-1)^3}} f''(x)$$

sign analysis of f''

numerator: $3x^2+1 > 0$

denominator: $(x^2-1)^3$

when is $(x^2-1)^3 > 0$? $\Leftrightarrow x^2-1 > 0$

$$x^2 > 1$$

$$x > 1 \text{ or } x < -1$$

f concave up

Conclusion: $\underbrace{f''(x) > 0}$ over $(-\infty, -1) \cup (1, \infty)$

$\underbrace{f''(x) < 0}$ over $(-1, 1)$

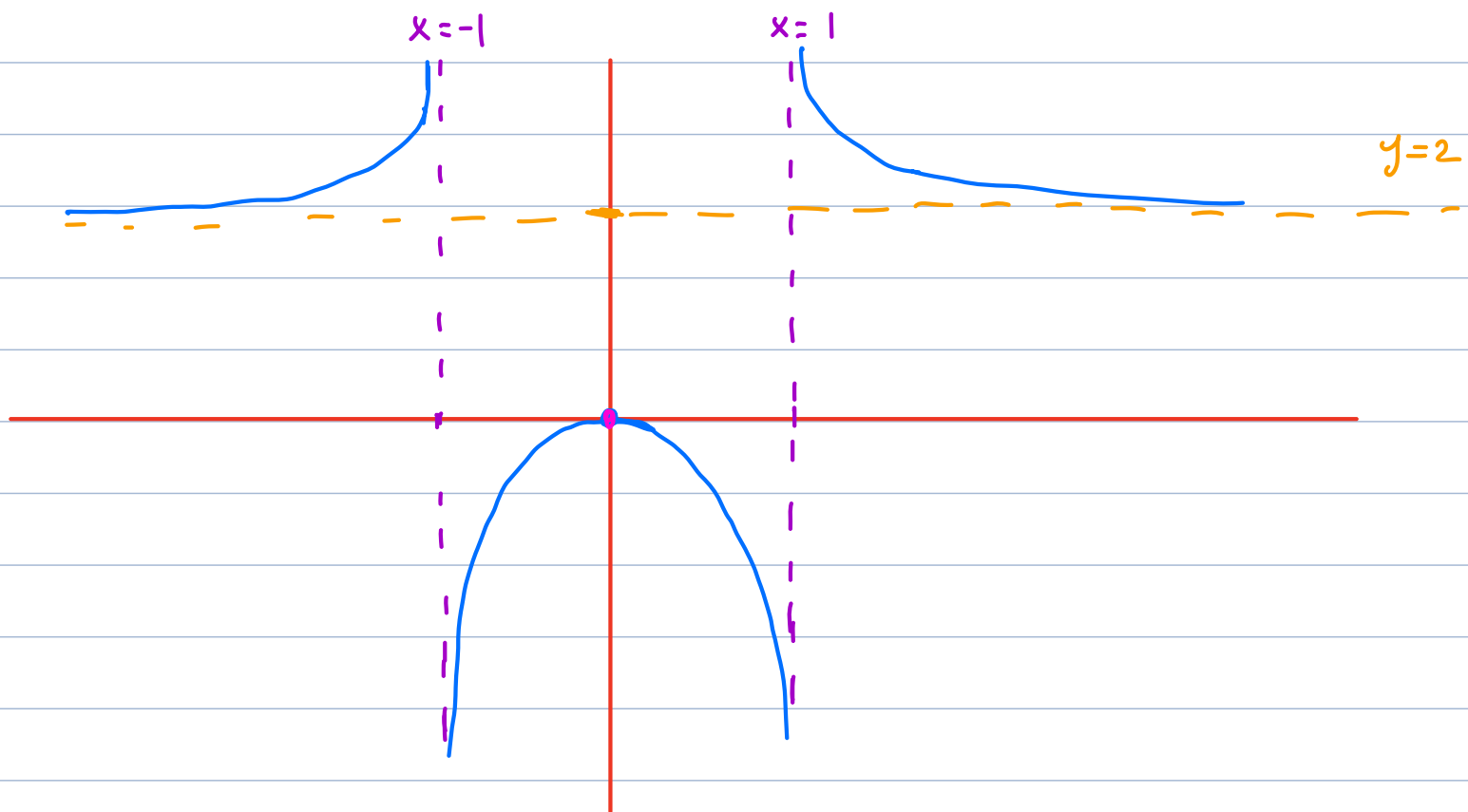
f concave down

inflection points? no inflection points

(concavity changes at ± 1 but function is undefined there so doesn't count)

sketch of function: (should reflect all information we found)

to sketch an even function, focus on positive x ,
then reflect over the y axis.



increasing
concave up

increasing
concave down

decreasing
concave down

decreasing,
concave up

Example sketch the graph of $f(x) = xe^x$

A domain: $(-\infty, \infty)$

B intercepts: y-intercept: $x=0: y = 0 \cdot e^0 = \underline{0}$.

x-intercept: $y=0: xe^x = 0 \Rightarrow \underline{x=0}$

C symmetry: no symmetry

$f(-x) = -xe^{-x}$ not equal to $f(x)$ or $-f(x)$

neither even nor odd

not periodic

D. Asymptotes: VA: none

HA: $f(x) = xe^x$



$$\lim_{x \rightarrow \infty} \underbrace{x}_{\downarrow \infty} \underbrace{e^x}_{\downarrow \infty} = " \infty \cdot \infty " = \infty$$

$$\lim_{x \rightarrow -\infty} \underbrace{x}_{\downarrow -\infty} \underbrace{e^x}_{\downarrow 0} = "0 \cdot \infty" \text{ indeterminate} \\ \Rightarrow \text{use L'Hospital's}$$

$$= \lim_{x \rightarrow -\infty} \frac{x}{e^{-x}} \stackrel{\text{L'Hospital's}}{=} \lim_{x \rightarrow -\infty} \frac{\frac{d}{dx}(x)}{\frac{d}{dx}(e^{-x})}$$

$$= \lim_{x \rightarrow -\infty} \frac{1}{-e^{-x}}$$

$$= \lim_{x \rightarrow -\infty} -e^x = \boxed{0}$$

$$\boxed{y=0 \text{ H.A.}}$$

E. increasing/decreasing

$$f(x) = xe^x$$

$$f'(x) = \underbrace{1 \cdot e^x + x \cdot e^x}_{\text{product rule}} = \underbrace{(1+x)}_{\uparrow} \underbrace{e^x}_{\text{always positive}}$$

sign analysis:

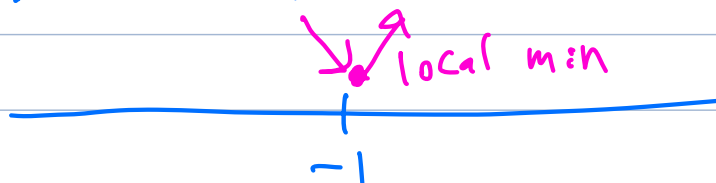
$$\underbrace{1+x > 0}_{\uparrow} \text{ when } \underbrace{x > -1}$$

$$f'(x) > 0 \text{ when } x > -1 \quad \text{increasing over } (-1, \infty)$$

$$f'(x) < 0 \text{ when } x < -1 \quad \text{decreasing over } (-\infty, -1)$$

$$f'(x) = 0 \text{ when } x = -1$$

F. local min/max



$$\text{local min value: } f(-1) = -1 \cdot e^{-1} = -\frac{1}{e}$$

$$f(x) = xe^x$$

G. concavity / inflection analyze f''

$$f'(x) = (1+x)e^x$$

$$f''(x) = \text{product rule}$$

$$1 \cdot e^x + (1+x) \cdot e^x$$

$$= \underbrace{(2+x)}_{\text{always positive}} e^x$$

$$2+x > 0$$

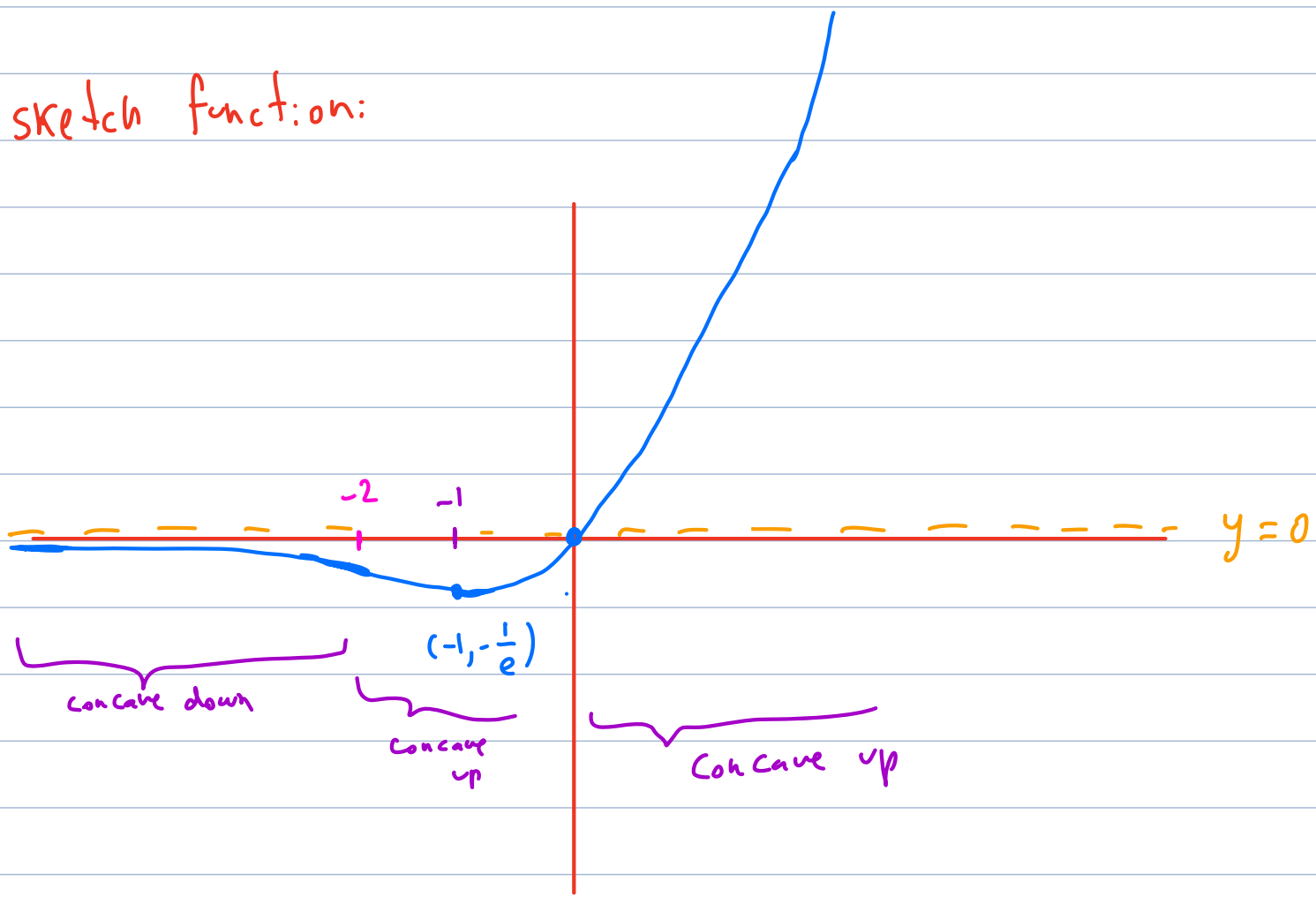
$$\text{when } x > -2$$

$f''(x) > 0$ when $x > -2$ } concave up over $(-2, \infty)$

$f''(x) < 0$ when $x < -2$ } concave down over $(-\infty, -2)$

$f''(x) = 0$ at $x = -2$ } inflection at -2

sketch function:



Example Sketch the graph of $y = \frac{\cos(x)}{2 + \sin(x)} = f(x)$

A domain: $2 + \sin(x) = 0$? never. domain = $(-\infty, \infty)$.

B intercepts.

y-intercept: $x=0$. $y = \frac{\cos(0)}{2 + \sin(0)} = \frac{1}{2}$

x-intercepts: $y=0$. $\frac{\cos(x)}{2 + \sin(x)} = 0 \Rightarrow \cos(x) = 0$
 $\Rightarrow x = \frac{\pi}{2} + n\pi$
 n whole number

C symmetry:

not even or odd

periodic, with period 2π

(if we shift graph horizontally by 2π it stays the same)

$$f(x + 2\pi) = \frac{\cos(x + 2\pi)}{2 + \sin(x + 2\pi)} = \frac{\cos(x)}{2 + \sin(x)} = f(x)$$

so can focus on $[0, 2\pi]$, get the rest of function by shifted copies of the function over $[0, 2\pi]$

D. Asymptotes:

VA: none (never dividing by 0). HA: none (periodic functions have no HA)

E. increasing/decreasing restricted to $[0, 2\pi]$

$$f(x) = \frac{\cos(x)}{2 + \sin(x)}$$

quotient
rule

$$f'(x) = \frac{\frac{d}{dx}(\cos x) (2 + \sin x) - \cos x \cdot \frac{d}{dx}(2 + \sin x)}{(2 + \sin x)^2}$$

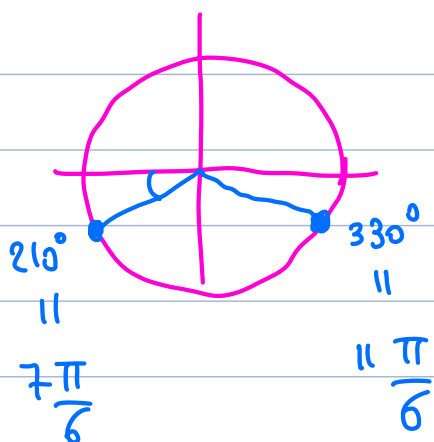
$$= \frac{-\sin x (2 + \sin x) - \cos x (\cos x)}{(2 + \sin x)^2}$$

$$= \frac{-2\sin x - \sin^2 x - \cos^2 x}{(2 + \sin x)^2}$$

$$= - \frac{2\sin x + \sin^2 x + \cos^2 x}{(2 + \sin x)^2} \quad \text{(trig identity)}$$

$$f'(x) = \underbrace{-}_{\text{neg}} \underbrace{\frac{2\sin x + 1}{(2 + \sin x)^2}}_{\text{positive}} \quad ?$$

Critical points: $f'(x) = 0 \Rightarrow 2\sin x + 1 = 0$



$$\Rightarrow \sin x = -\frac{1}{2}$$

$$\Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6}$$

critical points
over $[0, 2\pi]$

$$2\sin x + 1 > 0 \quad \text{if} \quad \sin x > -\frac{1}{2}$$

	0	$\frac{7\pi}{6}$	$\frac{11\pi}{6}$	2π
$2\sin x + 1$		+	-	+
$f'(x)$		-	+	-



increase over $(\frac{7\pi}{6}, \frac{11\pi}{6})$

decrease over $(0, \frac{7\pi}{6}) \cup (\frac{11\pi}{6}, 2\pi)$

F. local max/min

$\frac{7\pi}{6}$: local min

$\frac{11\pi}{6}$: local max.

G: skip concavity for this problem...

sketch the function:

these are all the same
by periodic behavior

